

MOHAMED HELMY

Android Software Engineer

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Summary

Android Software Engineer specializing in building modern, user-centric applications using Kotlin and Jetpack Compose. Experienced in Clean Architecture, MVI, and Kotlin Multiplatform, with a strong focus on scalable systems, offline-first solutions, and delivering high-quality, production-ready code.

WORK EXPERIENCE

Android Software Engineer, Areeb Technology

07/2025 – Present
Cairo, Egypt

- Leading the refactoring of a legacy multi-activity application to a modern Single Activity Architecture using Compose Navigation 3, improving stability and flow.
- Led research and proof-of-concept development for Kotlin Multiplatform adoption, presenting technical analysis to engineering leadership to drive cross-platform initiatives.
- Building core e-commerce modules including authentication and product management, utilizing GraphQL for efficient data synchronization.

Android Software Engineer, Freelance | POS & ERP Solution

05/2025 – 09/2025

- Engineered a production-ready Point of Sale (POS) application using Kotlin Multiplatform, delivering a unified codebase for Android tablets and Desktop.
- Designed a robust "Offline-First" architecture using Room and Supabase, implementing intelligent Delta Sync to handle transactional data seamlessly without internet connectivity.
- Built secure server-side RPC functions for critical stock transfers and inventory management, ensuring 100% data consistency across distributed devices.

Military Service, Egyptian Armed Forces

10/2023 – 12/2024

- Fulfilled mandatory national service, demonstrating adaptability, discipline, and the ability to work effectively under pressure in high-stress environments.

EDUCATION

B.Sc. in Computer Science, Benha University

09/2019 – 06/2023
GPA: 3.24

Graduation Project: Diabetes Health Indicators 📄 - Researched and developed a machine learning solution to predict diabetes. Analyzed a dataset of health metrics, engineered features, and evaluated multiple classification models, achieving high predictive accuracy.

SKILLS

Mobile Development:

- Android: Kotlin, Jetpack Compose, Clean Architecture, MVVM, MVI, Coroutines, Flow, Navigation (Nav2 & Compose Nav3).
- Kotlin Multiplatform (KMP): Compose Multiplatform, Ktor, Koin, Room, Multi-Module Architecture.
- iOS (Basics): Swift, SwiftUI, Xcode, SPM

Backend & Data:

- Supabase: Auth, Database, Realtime Sync, Edge Functions, Row-Level Security (RLS).
- API Integration: GraphQL, REST, Retrofit.

DevOps & Tools:

- CI/CD: GitHub Actions, Gradle.
- Testing: JUnit, Mockito/Mockery (Unit Testing), Compose UI Testing.
- Version Control: Git, GitHub.

PROJECTS

Safar, *Delivery Operations Platform (Kotlin Multiplatform)*

- Architecture: Built a scalable logistics platform for Android & Desktop using Compose Multiplatform and MVI, automating payroll and operations for delivery fleets.
- Backend Engineering: Developed Supabase Edge Functions to handle complex financial logic (salary/ deductions) and implemented Row-Level Security (RLS) for role-based data protection.
- Impact: Designed to reduce manual operational overhead by 90% through automated invoice tracking and real-time driver management.

hijri-date-picker [🔗](#), *Kotlin Multiplatform Library | Open-source*

Developed a production-ready Kotlin Multiplatform library for Hijri calendar date selection, supporting Android, iOS, and Desktop platforms. Actively maintained with comprehensive documentation and regular feature updates.

CrossCart [🔗](#)

Leveraged Compose Multiplatform to develop a single codebase for Android, iOS, and Desktop, significantly reducing development time. Implemented a clean, MVI architecture and real-time data synchronization with Supabase.

LocationReminder [🔗](#)

Engineered a reliable offline-first experience using Room for local data storage, ensuring users receive critical, location-based reminders without an internet connection.

IoT Environmental Monitor [🔗](#)

Engineered a full-stack IoT solution where an ESP8266 sensor streams real-time data to Firebase, displayed in a native Android app with a live dashboard and push notifications.

CERTIFICATIONS

Android Developer Track: Digital Egypt Builders Initiative (DEBI) | 2025

Android Application Developer Track: ITI | 2023

Advanced Android Nanodegree: Udacity & FWD | 2022

Machine Learning Specialization: DeepLearning.AI (Coursera) | 2022

CS50: Harvard University | 2021

Algorithmic Toolbox: Coursera | 2021